Assignment: Style Transfer

Implement and compare three image style-transfer techniques—**Neural Style Transfer (NST), Adaptive Instance Normalization (AdaIN), and Stable Diffusion–based Style Transfer**—to transfer the artistic style of **Jainul Abedin's paintings** and **SM Sultan's paintings** to the content images captured by your smartphones or cameras.

Tasks

1. Dataset Preparation

Use the dataset prepared by all students as the **content-image dataset**. Select at least **3 paintings by Jainul Abedin** as style/reference images.

2. Neural Style Transfer

Implement classical **VGG-19-based Neural Style Transfer (NST)**. For each content image, use a Jainul Abedin painting as the style image and generate a stylized image.

Use `conv4_2` for the content loss and `conv1_1`, `conv2_1`, `conv3_1`, `conv4_1`, and `conv5_1` for the style loss. Use Gram matrices to calculate the style loss. Optimize the generated image using gradient descent.

Explain the role of the VGG-19 network, content loss, style loss, Gram matrix, and image optimization.

3. Adaptive Instance Normalization

Implement **AdaIN** for arbitrary style transfer using the same content images and Jainul Abedin paintings.

Explain how AdaIN transfers style by aligning the channel-wise mean and standard deviation of content and style feature representations. Explain the roles of the pretrained encoder, AdaIN operation, decoder, content loss, and style loss.

4. Stable Diffusion–Based Style Transfer

Implement a **Stable Diffusion–based style-transfer method** using Jainul Abedin's paintings as style/reference images.

Generate stylized versions of the same content images used in NST and AdaIN. Experiment with appropriate parameters such as denoising strength, guidance scale, and inference steps.

Explain the role of the VAE, denoising network, conditioning mechanism, and diffusion process in generating the stylized images.

5. Comparative Evaluation

For the same content image and Jainul Abedin style image, generate three outputs using:

- Neural Style Transfer
- AdaIN
- Stable Diffusion

Compare the three methods based on:

1. Content preservation
2. Similarity to Jainul Abedin's painting style
3. Texture and color transfer
4. Visual/artistic quality
5. Processing time
6. Computational requirements
7. User controllability

Provide both **qualitative and quantitative comparisons**, where appropriate.

6. Discussion

Discuss the evolution of style-transfer techniques:

NST → AdaIN → Stable Diffusion–based Style Transfer

Explain how the fundamental style-transfer mechanism changes from **Gram-matrix-based feature correlations**, to **feature-statistics alignment**, to **diffusion-based generative modeling**.

Answer the following questions:

1. Which method preserves the content structure best?
2. Which method transfers Jainul Abedin's style most effectively?
3. Which method produces the highest visual quality?
4. Which method is fastest?
5. Which method requires the most computational resources?
6. Which method provides the greatest control over the generated image?
7. Why can Stable Diffusion modify the original content structure?
8. Why does classical NST require optimization for each content–style pair?
9. Why is AdaIN faster than classical NST?
10. Which method would you recommend for transferring Jainul Abedin's painting style to new images, and why?

Submission Requirements

Submit a report containing:

1. Introduction
2. Dataset description
3. Methodology
4. NST implementation and results
5. AdaIN implementation and results
6. Stable Diffusion implementation and results
7. Comparative analysis
8. Discussion
9. Conclusion
10. Source code
11. Generated images

Use the **same content images and Jainul Abedin paintings across all three methods** wherever possible so that the comparison is fair.